

Data Mining Projects – 2026

Each project requires to process one or several datasets by addressing one or several data mining tasks (e.g. feature selection/extraction, classification, clustering, association rule analysis, regression, timeseries forecasting, anomaly detection etc) by using any of the appropriate algorithms.

A project should contain:

- A **report** (around 6-8 pages) containing the description of the dataset, of the processing steps and of the results.

Structure of the report

- *Abstract* (2-3 paragraphs): briefly describe which are the objectives of the project and which are the main results.
- *Introduction*: describe the addressed problem, the motivation of choosing the dataset(s) and summarizes existing results on the same dataset(s) with clear references to state of the art publications (based on papers that used the same dataset); briefly presents the idea of the solution and how the report is structured.
- *Description of the dataset*: the characteristics of the dataset should be presented, including statistical analysis: no. of records, no. of attributes, types of attributes, distribution of attribute values (by type: histogram, mean, median, mode, standard deviation), data visualization (if applicable), percentage of missing values, degree of imbalance (if applicable).
- *Description of the processing flow*: describe the processing carried out, giving reasons for the selection of the used methods and details of implementation (own contributions should be emphasized)
- *Presentation of results*, comparison with those presented in other papers or on UCI Machine Learning/Kaggle (if applicable)
- *Conclusions*: brief presentation of the observations resulting from the implementation and application of the processing flow on the dataset;
- **Implementation of the processing workflow**: a Python notebook containing the processing steps applied to the dataset and comments related to the applied steps, the parameter values which have been used, values of the performance measures, comparisons between different methods (if it is the case), graphical illustrations of the data and/or results.
- **Slides** corresponding to an oral presentation of 10 minutes.
- **Remarks**:
 - Some project topics are more relevant for a MSc program (specified in the list) but there are no constraints – any student can select any project.
 - A project can be prepared in a team of 2 students. The report should clearly state the contribution of each one and both students should be involved in the oral presentation.
 - Each of the project components (report, implementation, oral presentation) will be evaluated and the project grade will be the average of the grades corresponding to the components.
 - AI generative tools can be used only for documentation.

List of proposed topics

1. Email Spam Detection (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/spambase>
- Task: Feature selection + classification
- Methods: PCA, Naive Bayes, SVM

2. Breast Cancer Diagnosis (AIDC)

- Dataset: [https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+\(diagnostic\)](https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+(diagnostic))
- Task: Feature selection + classification
- Methods: PCA, Decision Trees, SVM

3. Human Activity Recognition (ISR)

- Dataset:
<https://archive.ics.uci.edu/ml/datasets/human+activity+recognition+using+smartphones>
- Task: Classification
- Methods: kNN, MLP, Random Forest

4. Heart Disease Prediction (AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/heart+disease>
- Task: Feature selection + classification
- Methods: PCA, Naive Bayes, Decision Trees

5. Handwritten Digit Recognition (AIDC)

- Dataset:
<https://archive.ics.uci.edu/ml/datasets/optical+recognition+of+handwritten+digits>
- Task: Classification
- Methods: MLP, SVM, kNN

6. Customer Churn Prediction (BigData)

- Dataset: <https://www.kaggle.com/datasets/blatchar/telco-customer-churn>
- Task: Classification
- Methods: Random Forest, Boosting, SVM

7. Parkinson's Disease Detection (AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/parkinsons>
- Task: Feature selection + classification
- Methods: PCA, SVM, Random Forest

8. Ionosphere Signal Classification (AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/ionosphere>
- Task: Classification
- Methods: kNN, Naive Bayes, SVM

9. Phishing Website Detection (Big Data, Cybersecurity)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/phishing+websites>
- Task: Classification
- Methods: Random Forest, SVM

10. Adult Census Income Patterns (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/adult>
- Task: classification
- Methods: Decision Trees, SVM, Random Forest etc

11. News Article Clustering (BigData)

- Dataset: <https://www.kaggle.com/datasets/rmisra/news-category-dataset>
- Task: Clustering
- Methods: k-means, DBSCAN

12. Retail Customer Segmentation (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/online+retail>
- Task: Clustering
- Methods: k-means, Hierarchical

13. Image Color Quantization (BigData, AIDC)

- Dataset: <https://www.kaggle.com/datasets/crawford/emnist>
- Task: Clustering
- Methods: k-means

14. Social Network User Clustering (BigData)

- Dataset: <https://snap.stanford.edu/data/ego-Facebook.html>
- Task: Clustering
- Methods: DBSCAN, Hierarchical

15. Wholesale Customers Segmentation (AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/wholesale+customers>
- Task: Clustering

- Methods: k-means, DBSCAN

16. Seeds Dataset Clustering (AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/seeds>
- Task: Clustering
- Methods: Hierarchical, k-means

17. Yeast Protein Localization Clustering (BigData, AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/yeast>
- Task: Clustering
- Methods: DBSCAN, Hierarchical

18. Intelligent Log Monitoring (BigData, ISR, Cybersecurity)

- **Dataset:** HDFS Log Dataset – <https://github.com/logpai/loghub>
- **Task:** Clustering + anomaly detection
- **Methods:** k-means, DBSCAN, Hierarchical clustering

19. Energy Efficiency Prediction (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/energy+efficiency>
- Task: Regression
- Methods: MLP, SVR

20. Bike Sharing Demand Prediction (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/bike+sharing+dataset>
- Task: Regression
- Methods: Random Forest, Regression Trees

21. Concrete Strength Prediction (BigData, AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/concrete+compressive+strength>
- Task: Regression
- Methods: MLP, SVR

22. Air Quality Prediction (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/air+quality>
- Task: Regression
- Methods: RBF networks, Random Forest

23. Yacht Hydrodynamics Prediction (AIDC, ISR)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/yacht+hydrodynamics>
- Task: Regression
- Methods: MLP, RBF

24. Abalone Age Prediction (Big Data, AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/abalone>
- Task: Regression
- Methods: SVR, Random Forest

25. Online Retail Basket Analysis (BigData, ISR)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/online+retail>
- Task: Association rules
- Methods: FP-Growth

26. Movie Recommendation Patterns (BigData, ISR)

- Dataset: <https://grouplens.org/datasets/movielens/>
- Task: Association rules
- Methods: Apriori

27. Book Purchase Patterns (BigData)

- Dataset: <https://www.kaggle.com/datasets/saurabhbagchi/books-dataset>
- Task: Association rules
- Methods: FP-Growth

28. Stock Price Forecasting (BigData)

- Dataset: <https://www.kaggle.com/datasets/jacksoncrow/stock-market-dataset>
- Task: Forecasting
- Methods: Autoregressive models

29. Electricity Consumption Forecasting (BigData)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/electricityloadaddiagrams20112014>
- Task: Forecasting
- Methods: AR models

30. Weather Temperature Prediction (BigData)

- Dataset: <https://www.kaggle.com/datasets/sudalairajkumar/daily-temperature-of-major-cities>
- Task: Forecasting
- Methods: AR

31. Monthly Sunspot Activity Forecasting (BigData, AIDC)

- Dataset: <https://www.sidc.be/silso/datafiles>
- Task: Time series forecasting
- Methods: Autoregressive models

32. Atmospheric CO₂ Forecasting (BigData, AIDC)

- Dataset: <https://gml.noaa.gov/ccgg/trends/data.html>
- Task: Time series forecasting
- Methods: AR models

33. River Flow Time Series Prediction (BigData, AIDC)

- Dataset: <https://waterdata.usgs.gov/nwis>
- Task: Time series forecasting
- Methods: Autoregressive models

34. Intrusion Detection System (Cybersecurity)

- Dataset: <https://www.unb.ca/cic/datasets/ids.html>
- Task: Classification
- Methods: Random Forest, SVM

35. Sentiment Analysis (BigData, AIDC)

- Dataset: <https://ai.stanford.edu/~amaas/data/sentiment/>
- Task: Classification
- Methods: Naive Bayes, MLP

36. Network Traffic Clustering (Cybersecurity) – several datasets

- Dataset: <https://www.unb.ca/cic/datasets/index.html>
- Task: Clustering
- Methods: DBSCAN

37. Customer Lifetime Value Prediction (Big Data)

- Dataset: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
- Task: Regression
- Methods: Random Forest

38. Sales Forecasting (Big Data)

- Dataset: <https://www.kaggle.com/c/rossmann-store-sales>
- Task: Forecasting

- Methods: AR

39. Student Performance Prediction (AIDC)

- Dataset: <https://archive.ics.uci.edu/ml/datasets/student+performance>
- Task: Classification / regression
- Methods: Decision Trees, Random Forest

40. Anomaly Detection in Credit Data (Cybersecurity)

- Dataset: <https://www.kaggle.com/datasets/arjunbhasin2013/ccdata>
- Task: Clustering
- Methods: DBSCAN, k-means